

BODAPATI VISHNU SAI VARDHAN

Game Developer | XR Developer | Spatial Computing Builder | Technical Creative

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PROFILE

Game and XR developer pursuing an M.Eng in Game Design, Development and Innovation at Duke University, with hands-on experience building Meta Quest mixed reality tools, clinical VR rehabilitation interactions, Unity/Unreal game systems, spatial computing tools using 3D Gaussian Splatting, and AI-enabled web products. Strong focus on spatial interaction, gameplay systems, performance-aware XR implementation, UI clarity, and rapid prototyping from concept to playable demo.

EDUCATION

Duke University – Master of Engineering, Game Design, Development and Innovation Aug 2025 – Present

Pratt School of Engineering | Key coursework: Advanced Game Design, Unreal Engine Development, C++ Programming

University of Wollongong – Bachelor of Computer Science, Game and Mobile Development Sep 2021 – Sep 2024

Key coursework: Game Engine Essentials, VR/AR Game Development, Interactive Computer Graphics, Mobile Application Development, IT Project Management

AWARDS & HACKATHONS

1st Place: Best Spatial Reconstruction Project (by XGRIDS), 1st Place: Best App Running on Emulator (by PICO) & Runner-Up: Best Interactive World Experience (by World Labs) Jul 2026

CinemaScout, Worlds in Action Hack [02-LA]

- Won three awards at Worlds in Action Hack [02-LA] for a spatial cinematography copilot combining 3D Gaussian Splatting, Unity XR, and PICO deployment.

1st Place – Grand Winner, DesignXR Hackathon 2026

MR Blueprint – Mixed Reality Physics Sandbox

- Awarded Grand Winner for a Unity-based Meta Quest mixed reality sandbox enabling interactive 3D scene creation, drawing, editing, and physics simulation.

Winner – Honorable Mention, Team USA x Google Cloud Hackathon

Draft USA – The LA28 Momentum League | Devpost project | Project Link

- Built a Gemini-powered fan engagement app for drafting Team USA sports, Olympic-Paralympic pairings, and hometown hubs through transparent momentum scoring and AI-assisted roster guidance.

Top 50 Semifinalist – DevStudio 2026 by Logitech Hackathon

MR Blueprint – selected from a hackathon with 1,323 participants

- Advanced to the Top 50 semifinalist stage for a spatial computing prototype using Meta Quest, Unity XR, and Logitech MX Ink stylus interactions.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant – Duke Intelligent Interactive Internet of Things (I3T) Lab

Oct 2025 – Present

VR Gameplay & Research Developer | Duke University

- Develop VR-based rehabilitation gameplay for ICU patients, prioritizing seated/fixed-position comfort, accessibility, clear feedback, and constrained patient movement scenarios.
- Implement and iterate on Unity/Meta Quest gameplay mechanics, tutorial and calibration flows, interaction feedback, and repeated-session reliability for clinical research use.
- Optimize frame-rate stability, input responsiveness, and interaction smoothness to reduce motion discomfort risk in clinical VR settings.
- Collaborate with researchers and developers to translate clinical requirements into usable gameplay systems, testing workflows, and patient-friendly interaction design.
- Evaluate integration paths for Gaussian Splatting and AI-generated scenes inside Lumi to support richer, more realistic therapy environments.

Freelance Research Analyst – Singapore Gaming & Graphics Startup Ecosystem

Oct–Dec 2024

Exelvion IT Labs LLP

- Conducted market research across gaming, graphics, robotics, 3D modelling, VR/AR development, and artificial intelligence sectors.

- Compiled industry trends, startup performance metrics, investment activity, competitor strategies, and emerging technology opportunities for client-facing reports.
- Produced data-driven insights that supported strategic decisions, business development initiatives, and sector-specific opportunity evaluation.

SELECTED PROJECTS

CinemaScout: Spatial Cinematography Copilot

Jul 2026

Unity | C# | 3D Gaussian Splatting | OpenXR | PICO Unity Integration SDK | WebSpatial | Claude/GPT | Worlds in Action Hack [02-LA] | [Devpost](#)

- Built an immersive Unity application enabling filmmakers to explore 3D Gaussian Splat reconstructions of real locations and plan camera shots at real-world scale ahead of physical site visits.
- Implemented a virtual cinema camera system with adjustable focal length, field of view, aspect ratio, camera height, and clipping settings, plus spline based tracking shot playback previewable inside a VR headset.
- Contributed to an AI assisted shot recommendation feature and a WebSpatial “Mission Control” dashboard for organizing, comparing, and reviewing candidate viewpoints.
- Won 1st Place: Best Spatial Reconstruction Project (XGRIDS), 1st Place: Best App Running on Emulator (PICO), and Runner-Up: Best Interactive World Experience (World Labs).

MR Blueprint – Mixed Reality Physics Sandbox

Jan–Apr 2026

Unity | C# | Meta Quest 3/3S | OpenXR | XR Interaction Toolkit | Meta XR SDK | Logitech MX Ink SDK

- Built a Meta Quest mixed reality physics sandbox where users can spawn, edit, draw, and simulate interactive 3D physics scenes inside an XR workspace.
- Implemented core C# systems for object spawning, selection, transform manipulation, world-space XR UI, inspector panels, and Edit/Simulate Mode physics configuration.
- Contributed Logitech MX Ink Draw Mode and PhysicsLens features, including spatial drawing, pressure-sensitive strokes, live graphing, physics visualization, and snapshot/restore workflows.

Draft USA – AI Sports Draft Strategy Platform

May 2026

Next.js 14 | TypeScript | Tailwind CSS | Gemini API | Google Cloud Run | Docker | Recharts | Devpost

- Built a responsible AI fan engagement app that turns Team USA sport discovery into a fantasy-style draft experience focused on momentum, parity, hometown breadth, and underfollowed sports.
- Integrated Gemini-powered features for scout recommendations, roster analysis, parity coaching, leaderboard explanations, and Commissioner Recaps using server-side API routes.
- Designed user-facing dashboard flows for Draft Room, Roster Dashboard, Hometown Hubs, Momentum Score breakdowns, bonus unlocks, and league storytelling.
- Implemented responsible-language constraints that frame scores as fan discovery signals rather than medal predictions, betting signals, or athlete rankings.

Tower of Tricks – UI Programmer | Build & Debugging

Aug–Dec 2025

Unreal Engine | Blueprints | UMG | Packaging | QA

- Designed and implemented menus, HUD elements, settings/pause flows, and player-facing feedback systems for a playable Unreal game project.
- Debugged editor-vs-packaged build issues related to level loading, map inclusion, asset references, initialization order, and gameplay stability.
- Managed packaging configuration and performed systematic QA across builds to validate gameplay, UI behavior, and deployment readiness.

Hungry Owl – Sole Developer

Aug–Dec 2025

Playdate | Lua | Gameplay Systems | UI

- Designed and developed a complete Playdate game from scratch, owning gameplay, systems, UI, scoring, enemy behavior, difficulty progression, and game state logic.
- Integrated hardware-specific input using the Playdate crank and buttons, mapping physical controls to responsive in-game aiming and interaction systems.

Overpriced – Game Jam

Nov 2025

Gameplay Programmer | Audio Programmer | Unreal Engine

- Implemented the core audio integration pipeline, including sound cue setup, event triggering, and contextual audio feedback tied to gameplay interactions.
- Programmed gameplay interaction logic supporting player-environment connectivity, puzzle flow, level progression, and responsive feedback states.

The Misadventures of Skylar Knight – Game Jam

Jan 2026

Gameplay Programmer | Audio Engineer | UI Developer | Unreal Engine

- Implemented player mechanics, interactions, core game logic, UI/audio feedback, 3D-to-2D gameplay adaptation, build testing, packaging, and version control support.

Meteor Mayhem – Hand Gesture-Based Game

Python | Computer Vision | Unity | C#

- Developed a gesture-controlled Unity game using Python-based real-time hand gesture recognition to drive in-game actions and natural interaction flow.

Additional Unity Academic Prototypes

FPS prototype | VR Maze | Interactable VR Room

- Built Unity prototypes covering first-person mechanics, enemy/weapon systems, VR locomotion, spatial audio, haptic feedback, object interaction, and physics-based manipulation.

TECHNICAL SKILLS

Engines & XR: Unity, Unreal Engine, Meta Quest 3/3S, XR Interaction Toolkit, OpenXR, Meta XR SDK, Logitech MX Ink SDK, PICO Unity Integration SDK, 3D Gaussian Splatting, WebSpatial, URP, world-space XR UI, interaction design

Programming: C#, C++, C, Python, JavaScript, TypeScript, Java, Lua, SQL/MySQL

Gameplay & Systems: Gameplay programming, physics interactions, UI/HUD systems, audio implementation, game states, input systems, build packaging, QA/debugging, performance optimization

Web, Cloud & AI: React, Next.js, Tailwind CSS, Gemini API, Claude/GPT APIs, Google Cloud Run, Docker, server-side API routes, prompt/agent constraints, responsible AI design

Tools: Git/GitHub, Android Studio, VS Code, Netlify, Formspree, basic video/editing workflows, documentation, technical research

CERTIFICATIONS, LEADERSHIP & LANGUAGES

Certifications: Young AI Pro Training Program - Advanced AI & LLM Techniques (2024); Game Development With Unity (2020, MAGES Institute of Excellence); Professional Remote Pilot License Program (2019, Ace Aviation); UI Design Course (2020, Udemy); User Experience (UX): The Ultimate Guide to Usability and UX (2020, Udemy)

Leadership: Entrepreneur Society Captain – led fundraising and supported event organization (2018)

Languages: English (fluent), Telugu, Hindi